02

CHEMISTRY

After analysing the previous year question papers, we have seen that 40-50 questions are asked from science section, out of these 15-18 questions are asked from Chemistry. From chemistry section, around 2-3 questions are asked from matter, man-made materials, 1-2 questions from atomic structure, electrochemistry, radioactivity, acid base and 4-6 questions from inorganic, environment and its pollution and 1-2 question each from chemical bonding and redox reaction gas law and solution, surface chemistry, chemical thermodynamics, organic chemistry etc.

MATTER

As we look at our surroundings, we see a large variety of things with different shapes, sizes and textures. Everything in this universe is made up of material, which scientists have named "matter".

MATTER

Anything that has mass, occupies space and can be felt by our one or more sense organs is called matter. Matter is found in five physical states-solid, liquid, gas, plasma and Bose-Einstein condensate. Out of which, three states, i.e. solid, liquid and gas are more common.

- **Solid** They have a fixed shape, volume and high density. The particles in solids are (closely) packed and held in rigid positions. They can be true or crystalline, (i.e. having ordered arrangement of constituent particles to a larger distance) like metals or amorphous or pseudo solids, (i.e. having short range arrangement of constituents particles) like non-metals, glass, most of the polymers, etc.
- **Liquid** They have a fixed volume but no fixed shape and have moderate to high densities. The particles in liquids are loosely packed and free to move. Their thermal conductivity decreases with rise in temperature with the exception of water.
- **Gas** They have neither a fixed shape nor a fixed volume and have very low density. The particles in gas are widely spaced apart and uniformly distributed in the container.

Two More States of Matter

- **Plasma** The state consists of super energetic and super excited particles which are in the form of ionised gases. Due to the presence of plasma, the sun and stars glow.
- **Bose-Einstein Condensate** (BEC) They are formed by cooling a gas of extremely low density to super low temperatures.

Properties of Matter

The properties of matter are as follow :

(i) **Density** It is the measure of relative heaviness of objects having constant volume and is defined as mass per unit volume. Mathematically, it can be written as :

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \text{g /mL}$$

At 4°C, density of water is maximum while its specific volume is minimum.

Gases possess very low density.

(ii) **Compression** Gases are more compressible because of the presence of large intermolecular spaces while solids are least compressible. Compressibility of liquids lies in between of the two other states.

(iii) **Thermal Expansion** It is minimum in case of solids but maximum in case of gases because intermolecular force of attraction is maximum in solids but minimum in gases.

Change of States of Matter

This is phenomenon of change of matter from one state to another and come back to original state by altering the original temperature and pressure.

(i) **Boiling Point** The process of conversion of liquid into vapours by heating is called boiling and the temperature at which the vapour pressure of a liquid becomes equal to the atmospheric pressure is called boiling point. Boiling point increases in the presence of impurity.

At high altitude, boiling point decreases because atmospheric pressure is low. Hence, food takes more time for cooking at higher altitudes. Similarly, in pressure cooker, food cooks early due to elevation in boiling point.

(ii) **Melting Point** The process of conversion of a solid into liquid by heating is called melting and the temperature at which a solid state to convert into its liquid state is called melting point.

(iii) **Sublimation** It is the process of heating a solid, so that it converts directly into gas. e.g. naphthalene, camphor.

(iv) **Freezing** It is the process of conversion of a liquid into solid at its freezing point and melting is the process of conversion of a solid into liquid at its melting point.

(v) **Evaporation** It is a process of conversion of a liquid into vapour at any temperature, while boiling is the conversion of a liquid into vapour at its boiling point. Evaporation causes cooling as the liquid takes energy for evaporation from surroundings.

(vi) **Latent Heat** When heat is supplied to a body either at its melting point or boiling point, the temperature of the body does not change. In this condition, heat supplied to the body is used up in changing its state and is called the latent heat.

$$\text{Latent heat } (L) = \frac{\text{Quantity of heat } (Q)}{\text{Mass } (m)}.$$

Its SI unit is Jkg^{-1}.

CLASSIFICATION OF MATTER

On the basis of chemical composition, matter is classified in the following manner:

Elements

The term element was first used by Robert Boyle in 1661. These are the simplest form of matter and therefore, cannot be split into simpler substances by any physical or chemical methods, e.g. sodium (Na), iron(Fe), mercury (Hg), etc. There are 118 elements known at present, out of which 98 occur in nature, while the remaining elements have been prepared artificially.

Compounds

These are made up of two or more elements combined in a fixed ratio. A compound cannot be separated into its components by physical methods.

- The properties of a compound are entirely different from those of its constituent elements.
- A compound has a fixed melting point, boiling point. A compound is a homogeneous substance. e.g. water, ammonia, sugar, etc.

Mixtures

A mixture is a substance which is made up of two or more elements or compounds, chemically combined together in any ratio (so no definite formula).

These can be separated into its constituents by the physical methods. A mixture does not have a fixed melting point, boiling point and shows the properties of its constituents.

There are two types of mixtures:

1. **Homogeneous Mixtures** In this type of mixture, the constituents are uniformly distributed through out. e.g. salt solution, sugar solution, air etc. These are also called true solutions.
2. **Heterogeneous Mixtures** In this type of mixture constituent does not have uniform composition. They may or may not settle on standing for a long time. They exhibit Tyndall effect. e.g. colloidal solutions, mixture of salt and sugar, iodised salt (mixture of potassium iodide and common salt) etc.

PHYSICAL AND CHEMICAL CHANGES

In physical changes, only the physical properties of matter like colour, hardness, density, etc., changes while the chemical properties and composition remain the same.

Crystallisation, boiling, dissolution of salt and sugar, vaporisation, melting of ice etc., are examples of physical change.

In chemical changes, the chemical composition as well as chemical properties of the matter changes and a new substance is formed. Burning of any substance, photosynthesis, ripening of fruits, etc., are examples of chemical change. Physical changes are reversible (i.e. can be reversed to obtain the original substance) while chemical changes are irreversible.

Laws of Chemical Combination

Combination of elements/atoms to form compounds is governed by following basic laws:

(i) **Law of Conservation of Mass** This law was put forth by **Antoine Lavoisier in 1789**. It states that, matter can neither be created nor destroyed but can be converted from one form to another.

In a chemical reaction, total mass of reactants is equal to total mass of products. In terms of energy, it is called law of conservation of energy.

(ii) **Law of Definite Proportions** A pure compound always contains same elements combined in same proportions by mass, whatever be its source. This is called law of definite proportions by weight and is given by **Joseph Proust**, e.g. water obtained from any source like tap water, well water, etc., always contain hydrogen and oxygen in 1:8 by mass.

(iii) **Law of Multiple Proportions** This law was proposed by **John Dalton in** 1803 . It states that when the fixed mass of an element combines with different masses of another element, then the masses of other elements (in two or more compounds) are in the ratio of small whole numbers..

e.g. H_2O 2 g : 16 g

H_2O_2 2 g : 32 g

or 1 g : 32 g.

Thus, ratio of O combining with same mass of H = 1 : 2.

(iv) **Gay Lussac's Law of Gaseous Volumes**The volume of reactants and products in large number of chemical reactions are related to each other by small integers, provided the volumes are measured at same temperature and pressure. These statements are considered as the law of definite proportions by volume given by Gay-Lussac.

MOLECULES

The term molecule was given by Avogadro. It is the smallest particle of a compound which can exist in free state. All the properties of a compound depend on its constituent molecules. Molecules of a compound contain two or more different types of atoms in a fixed ratio. e.g. ammonia (NH_3), water (H_2O), etc.

- **Molecular mass** is the mass of a molecule. It is an additive property and is calculated by adding the atomic masses of total atoms present in a molecule.

  e.g. $NH_3 = 14 + (1) \times 3 = 17$.

- **Equivalent weight** is obtained by dividing molecular or atomic mass by valency.

$$\text{Equivalent weight} = \frac{\text{Molecular mass or atomic mass}}{\text{Valency}}$$

(i.e. equivalent weight is affected by change in valency)

Mole Concept

The number of molecules present in 12 g of C-12 atom is called one mole. 1 mol $= 6.022 \times 10^{23}$ = Avogadro's number

- Number of moles $= \dfrac{\text{Mass (in g)}}{\text{Atomic or molecular mass (g mol}^{-1})}$

$$1 \text{ mol of atom} = 6.022 \times 10^{23} \text{ atoms}$$

$$1 \text{ mol of molecules} = 6.022 \times 10^{23} \text{ molecules}$$

- **Atomic mass** is the relative mass as compared with an atom of C-12 and is expressed in amu.

  (1 amu $= 1.6 \times 10^{-24}$ g).

ATOMIC STRUCTURE

Atoms are the basis of chemistry, even they are the basis for everything in the universe. **Maharishi Kanad** was one of the first person to propose that matter is made up of every small particles called **parmanu**. **John Dalton** called the particles by the name of atom.

DALTON'S ATOMIC THEORY

In 1808, **John Dalton** gave a theory, called atomic theory of matter which is based upon laws of chemical combination.

According to this theory,

- Atom is the smallest indivisible particle of an element, i.e. it can neither be created nor be destroyed.
- Atoms of different elements differ in mass, size and chemical properties.
- Atoms of same or different elements combine together to form compound or molecule. The 'number' and 'kinds' of atoms in a given molecule is fixed.
- Atoms of the same elements can combine in more than one ratio to form different compounds.

Atom and Molecule

Atom is the smallest particle of the element that can exist independently and retain all its capable of independent chemical reactions. Atoms are made up of subatomic particles like electrons, protons, neutrons, meson, positron, etc.

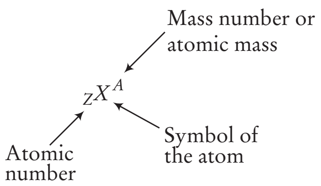
A molecule is the smallest particle of an element or a compound existence under ordinary conditions.

Properties of Atom

The properties of atom are as follow:

- **Atomic Number** For an element, it is the number of protons present inside the nucleus of its atom.
  Atomic number (Z) = number of protons
  = number of electrons
  (in case of neutral atom)
- **Mass Number** It is the total number of protons and neutrons (i.e nucleons) present in one atom of an element.
  Mass number (A)
  = number of protons + number of neutrons
  = atomic number + number of neutrons
  = number of electrons + number of neutrons

Representation of an Atom



Discovery of Subatomic Particles

The subatomic particles are discovered in many ways:

Cathode Rays

These rays travel in straight line and cast shadows of metallic object placed in their path. In the presence of electrical or magnetic field, the behaviour of cathode rays are similar to that of negatively charged particles, called **electrons.**

The first precise measurement of the charge on the electron was made by **Robert A. Millikan** in 1909.

Anode Rays

In 1886, **Engen Goldstein** discovered the canal rays. These rays travel in straight line and can also produce mechanical effects. These are positively charged, i.e. made up of **protons**.

Properties of Subatomic Particles

The properties of subatomic particles are as follow :

- **Electron** ($_{-1}e^0$) It is negatively charged particle. It was discovered by J. J. Thomson in 1897.
  $$\text{Charge on electron} = -1.6 \times 10^{-19}\text{C}$$
  $$\text{Mass of electron} = 9.1 \times 10^{-31}\text{g}$$
  The magnitude of negative charge on electron was determined by Millikan in his oil drop experiment.
- **Proton** ($_1\text{H}^1$) It was discovered by E. Goldstein. It is positively charged particle.
  $$\text{Charge on proton} = 1.6 \times 10^{-19}\text{C}$$
  $$\text{Mass of proton } (_1\text{H}^1) = 1.6 \times 10^{-27}\text{kg}$$
- **Neutron** ($_0n^1$) It was discovered by Chadwick. It has neutral particle, i.e having no charge.
  $$\text{Mass of neutron} = 1.6 \times 10^{-27}\text{kg}$$
  Hydrogen and protium is the only atom that does not possess neutrons.

Subatomic Particles

Particle	*Discoverer*	*Charge*	*Mass*
Electron	Thomson	–	1
Proton	Rutherford	+	1836
Neutron	Chadwick	0	1836
Meson	Yukawa	+, 0, –	273.8
Positron	Anderson	+	1
Neutrino	Fermi	0	20.04

NUCLEUS

Atom consists of a heavy and positively charged part at its centre, called the nucleus (dia = 10^{-4}m). Nucleus was discovered by Rutherford by α-particle scattering experiment. The negatively charged electrons revolve around the nucleus in well defined orbits. A nucleus consists of positively charged protons and electrically neutral neutrons.

Planck's Quantum Theory

(Particle Nature of Electromagnetic Radiation)

The radiant energy which is emited or absorbed discontinovsly in the form of small discrete packets of energy known as quantum and in care of light, the quantum of energy is called photon.

$$E = h\nu \qquad (c = \nu\lambda)$$
$$E = \frac{hc}{\lambda}$$

where, h = Planck's constant = 6.63×10^{-34} J-s

E = Energy of photon or quantum

If n is the number of quanta of a particular frequency and E_T be the total energy, then $E_T = nh\nu$. The energy possessed by one mole of quanta (or photon), i.e. Avogadro's number (N_0) of quanta, is called one Einstein of energy i.e., 1 Einstein of energy $(E) = N_0 h\nu = N_0 \dfrac{hc}{\lambda}$

Different Atomic Species

(*i*) **Isotopes** Atoms of the same element having the same atomic number (Z) but different mass number (A) are called isotopes,
e.g. $_1H^1$ (protium), $_1H^2$ (deuterium or heavy hydrogen) and $_1H^3$ (tritium) are isotopes of hydrogen.
$_6C^{12}$, $_6C^{13}$, $_6C^{14}$ are isotopes of carbon.
$_8O^{16}$, $_8O^{17}$, $_8O^{18}$ are isotopes of oxygen.
$_{10}Ne^{20}$, $_{10}Ne^{21}$, $_{10}Ne^{22}$ are isotopes of neon.

(*ii*) **Isobars** Atoms of different elements having the same mass number (A) but different atomic number (Z) are called isobars, e.g. $_1H^3$ and $_2He^3$; $_{18}Ar^{40}$, $_{19}K^{40}$ and $_{20}Ca^{40}$; $_{52}Te^{130}$, $_{56}Ba^{130}$ and $_{54}Xe^{130}$, etc.

(*iii*) **Isoelectronic Species** Species having the same number of electrons but different nuclear charge are known as isoelectronic species. They also have same bond order. e.g. Mg^{2+} and Na^+, etc., are isoelectronic species as both have 10 electrons.

(*iv*) **Isotones** These have the same number of neutrons. e.g. $_6C^{14}$, $_7N^{15}$.

Bohr-Bury Scheme

It shows the distribution of electrons in different shells, i.e. orbitals or paths of different and definite energies in which the electrons revolve. According to this scheme,

- A shell can have a maximum of $2n^2$ electrons (where, n = number of shells). e.g. the maximum number of electrons in
  K (or first) shell $= 2n^2 = 2(1)^2 = 2$
  L (or second) shell $= 2n^2 = 2(2)^2 = 8$ and so on.
- The outermost shell cannot have more than 8 electrons.
- Electrons enter in the new shell only after filling the previous one completely.

Quantum Numbers

Quantum numbers are just like address of electrons. There are four types of quantum numbers which are given below

(*i*) Principal quantum number $(n) = 1, 2, 3, 4, \ldots$

(*ii*) Azimuthal quantum number $(l) = 0$ to $n - 1$ for a given value of n.

(*iii*) Magnetic quantum number (m or m_l) $= -1$ to $+1$ including '0' for a given value of m.

(*iv*) Spin quantum number (s or m_s) $= +\dfrac{1}{2}, -\dfrac{1}{2}$ for a given value of m.

Electronic Configuration

It is the arrangement of electrons in various shells, sub-shells and orbitals in an atom. It is written as 2,8,8,18,32.
e.g. the electronic configuration of

	K	L	M	N
Sodium (Na_{11})	2,	8,	1,	0
Calcium (Ca_{20})	2,	8,	8,	2

RADIOACTIVITY

All heavy elements and a few of lighter elements have naturally occuring isotopes, which possess the property of radioactivity.

These isotopes have unstable nuclei and attain stability through the phenomenon of radioactivity.

The unstability results in the emission of a complex type of powerful radiations known as alpha (α), beta (β) and gamma (γ) rays.

RADIOACTIVITY

Radioactivity was discovered by a French physicist **Henri Becquerel** in 1896. However, the term radioactivity was given by Marie Curie, the scientist who got Nobel Prize twice (for physics and chemistry).

The spontaneous emission of invisible radiations by disintegration of heavy elements into comparatively lighter elements is called **radioactivity**. The invisible rays emitted by radioactive elements consists of the following particles:

(*i*) **Alpha (α) particles**, i.e. $_2He^4$ (+ 2 unit charge and mass four units) They are deflected towards negative plate in the electric and magnetic field and have very high ionising power.

$$_ZX^A \xrightarrow{\alpha\text{- particle}} {}_{Z-2}Y^{A-4}$$

(*ii*) **Beta (β) particles**, i.e. electrons (−1 charge and zero mass) They are deflected towards positive plate in the electric and magnetic field.

$$_ZX^A \xrightarrow{\beta-\text{particle}} {}_{Z+1}Y^A$$